BIOLOGICAL TRANSHUMANISM AS ENHANCED NATURE:
A CALL FOR FUNDING RESEARCH ON HEALTHY LONGEVITY

AT THE OUTSET (unnecessary?)

Adaptation is not banal: to adapt means to change either the environment or oneself in order to transform a harsh environment into an optimal environment. That's how the thermophile bacteria succeeded in growing optimally in boiling water (and freezing at room temperature). Once such hard work of adaptation is successfully done, then, for the winner, it is best that nothing changes. The winner would like to freeze his victory!
Thus, the conservativism of the dominant social mainstreams is understandable. They competed with the previous mainstream by better adapting to the changing environment or by modifying the social environment to suit its own success, which means the privileged access to common resources. Biomedical research establishment is currently such an adapted winner that can allow itself the luxury of becoming mentally lazy, hence less fit and easier to dethrone. Here is a gentle attempt to do so.

A VISION AND A GOAL

Recent scientific insights and common sense lead us to believe that diseases are not unavoidable and death is not indispensable.
By now, there is a solid biological basis for claiming that we could all, or nearly all, live healthy productive life up to 120 years - without changing human biology. While being as human as other people, fellow supercentenarians demonstrate the reality of this claim. Their longevity and the quality of life are as enviable as the quality of their death (supercentenarians generally do not suffer from long lasting malignant or degenerative diseases). Extension of lifespan beyond 120 years would require well informed changes in human biology and we are not yet there.

By working in good health to the age of over 90 would harness the advantages of wisdom and experience. Old age and wisdom can serve societal harmony, or homeostasis, by reducing metastatic growth of certain human activities at the expense of other societal needs and commodities (e.g., health, education, security and conquest of happiness). Importantly, long productive life and short unproductive period would allow for much reduced rate (down to zero) of human population growth and hence ease many ecological and economic problems.

Technological transhumanism versus biological transhumanism*: Fundamentally, through biological transhumanism we enhance our species while through techno or cyber transhumanism we diminish human species (quote Z. Djermanovic). Transhumanism was conceived by its spiritual fathers as a panoply of electronic and chemical prostheses with the purpose of extending human life. But, in extreme, cyber transhumanism leaves human biological body as a banal chassis for standardized prostheses. Then, it is not clear why the chassis needs to be biological? What would happen to variation and selection for further*

evolution? Without our biological self, what would remain of human civilization without emotion, talent, sense of harmony, esthetics, ethics, etc.?

In contrast, biological transhumanism and well-planed symbioses with new commensals (again Djermanovic) maximizes what we already are in our unicity, or singularity, assuring further civilizational growth by building upon the past of human evolution. In the face of fast evolving human culture, in particular science, biological diseases and death are a shame. The question is how can we transgress our biological limits while remaining human? Actual supercentenarians provide a realistic model to look forward.

THE STATE OF ART: BIOMEDICAL RESEARCH IS ON A WRONG TRACK

Malignant and degenerative age-related diseases account for about 90% of deaths in economically developed countries. Over the last half century, there was a monumental world-wide investment (in trillions of dollars, engaging millions of MDs, PhDs and technicians) in mission-oriented research to mitigate deadly age-related diseases. Yet, they are still incurable. Such track record would forever ruin the career of a scientist but not the future of a failing but powerful research establishment bulimic for funds to develop new "cutting edge technologies". The diagnostic is that there is a flagrant lack of strong and productive scientific concepts.

Evidently, at its outset, modern biomedical research took a mainstream approach that is inadequate for the task. Moreover, it caused a collateral damage by preventing the funding of alternative approaches. In spite of dismal success, funding agencies and even the leading scientific journals collaborated in creating a single-minded monopoly.

Common sense would have predicted that the studies of sick people will teach nothing useful about the cause and mitigation of diseases. Such research revealed only, at the molecular level, a large repertoire of biological damage done by the diseases and nick named the biomarkers. There was no reason to expect that targeting the drugs to such biomarkers way downstream of the disease initiation (consequences of the consequences…) can reverse the disease. Such "smart drugs" should not and did not work.

Instead of studying millions of sick people (who merit the best care), common sense would have chosen a deep study of the biology of few (even one would do) of healthy supercentenarians with bad habits and risky lifestyles. Such biology would teach what does it take to become resistant simultaneously to all of over one hundred of malignant and degenerative diseases. In other words, we should have explored what we want (health and resilience) rather than what we don't want (disease).

WHY IS MITIGATION OF ALL AGE-RELATED DISEASES POSSIBLE?

It is highly unlikely that the increasing numbers of relatively rare centenarians is due to chance assembly in single genomes of genetic resistances to each of hundreds of deadly

diseases. (Only a long-lasting planed genetic outcrossing of sensitivities and inbreeding, like with pure-bread dogs and horses, could increase the probability of such resistance).
It is more likely that very diverse age-related diseases - both malignant and degenerative - share a common cause, or few common causes. The slope of the increase in the incidence of such numerous diseases with age is similar to the that of death (all increasing with about 5th power of time) displaying a likely common biological clock. Perhaps all these diseases and death share also a common cause? In such a case, counterintuitively, by acting upon the common cause, mitigating simultaneously all of couple of hundreds of age-related diseases would be simpler than mitigating only one of them. That would be a way of turning the entire human population into fit centenarians!

THE BASICS OF LIFE

Brief description of the role of genes and proteins. Proteins run all life functions, either directly or indirectly by synthesis, maintenance, repair and renewal of all biomolecules, including proteins, RNA and DNA. The most relevant and irreversible protein damage (carbonylation) is caused by oxygen free radicals (called ROS), resulting in protein oxidation, or simply - corrosion!

WHAT COULD BE THE COMMON CAUSE OF AGE-RELATED DISEASES?

We have shown that the most likely common cause that progressively ruins life's functions is damage to proteins - life's molecular nano-machines.

The most robust creatures - from bacteria to animals - protect specifically their proteins against corrosion. The degree of robustness (e.g., resistance to radiation, toxic chemicals and desiccation) is perfectly correlated with the efficiency of protein protection. The undamaged proteome can repair extreme damage to DNA and replace other damaged molecules by new ones. We established beyond doubt that the chemistry of bacterial and animal cell death by radiation is protein oxidation. Earl Stadtman and colleagues showed at the end of 1980-ies that lifespan correlates inversely with protein oxidation across the species from tiny worm to man. They suggested a reasonable causal relationship between irreversible oxidative protein carbonylation and death, but the arbitrary hypothesis of protein carbonylation as mere "biomarker" prevailed. Now, we know that it's causal, at least for bacterial cells.

Proteins of aerobic organisms (which produce ROS) are intrinsically very resistant to oxidation - that's why we can live hundred years. But, even subtle mistakes (e.g., substitution of only one amino acid) occurring in the protein-producing assembly line (called ribosomes) or benign protein polymorphisms (called silent mutations) can ruin protein resistance to oxidation. We found that those silent mutations that predispose a person to an age-related disease regularly predispose the relevant protein to oxidative damage. The severity of predisposition to Parkinson's or ALS disease is quantitatively mirrored by the predisposition of the relevant protein (synuclein or SOD1, respectively) to oxidative damage. This identifies the nature of inborn predispositions to diseases whose onset will happen several decades later. (The one-year long silence of the biomed establishment to

these publications sent to some prominent experts becomes loud and instigates thoughts about professional ethics.)

Late Susan Lindquist pioneered brilliant discoveries that numerous silent mutations (polymorphisms) in proteins are “silent”, without consequence, because of the activity of special proteins, called chaperones (like the experienced ladies on French courts watching over the virginity of noble young girls). Chaperones are the doctors of other proteins that are “sick” since misfolded due to a benign mutation or to a random mistake in their production. Oxidation of a protein destabilized by a silent mutation will fix and paralyze the aberrant protein structure leading to its malfunction, or worst - a toxic function that can kill the cell. Thus, only if chaperone comes first to a misfolded protein, it will correct its structure into a normal oxidation-resistant state. Thus, in acting upon misfolded proteins, refolding by chaperones and oxidation by ROS are competitive antagonists.
Hence, it appears that prevention and even attenuation of diseases is possible - without involving questionable efficacy of the correction of inborn and acquired mutations - at the level of the manifestation of mutations (called phenotype, such as a disease). Since age-related diseases are predisposed by mutations that are inconsequential (silent) for several first decades of life, the protection of fragile proteins against oxidation should keep the “silence” (attenuated state) of the predisposing mutation for indeterminate time.

MISSION

The lesson learned from (i) extremely resistant bacteria and animals and (ii) the nature of mutations predisposing to neurodegenerative diseases is that providing a protein protecting shield by antioxidant small-molecule chemical chaperones should keep disease-predisposing mutations in extended latency and could also slow down, even reverse, the phenotypes of aging caused by protein oxidation.
Such shield would keep the proteome in the state of cellular health similar to early decades of life, i.e., as long as the protective shield is effective, the initiation of disease should be largely avoided and what is avoided should stay in an attenuated state.

We have identified in MedILS the first described small-molecule pigmented antioxidant chaperone in an extremely radiation resistant bacterium that fell with snow from the atmosphere or stratosphere, highly irradiated by UV light in a desiccated form (i.e., “clinically dead”!) and “resurrected” in the laboratory. Many more of such molecules can be expected to be present in extremely resistant bacteria, archaea, plants and animals. The task is to look for chemical chaperone antioxidant molecules in robust organisms, identify them and study their biological properties. The mission is the mitigation of all deadly diseases.

IMPLEMENTATION OF THE PROJECT

The research leading to several basic concepts underlying biological robustness led to the identification of the causes and to a general understanding of the consequences of aging and age-related diseases. Two discoveries show a potential for promoting health and longevity via:

(a) Prevention of initiation of pathologies by the protection of proteins against oxidative damage
(b) Improving tissue homeostasis by enhancing cellular parabiosis that compensates the emerging cellular defects (recessive mutations and epigenetic gene silencing)
(c) Attenuation of already manifested diseases by two strategies: (i) enhancing cellular parabiosis between sick and healthy cells and (ii) allowing sick cells to start producing intact proteins by proteome renewal (synthesis and turnover) in the presence of chemical antioxidant chaperones. These approaches are expected to reverse the diseases back to health.

A well-planed approach to use best-adapted tests to screen for stimulation and maintenance of cellular parabiosis and for antioxidant chemical chaperones from well-selected biological sources will be elaborated.